Branch/Class/Batch: AI&DS/TE/A1
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EXPERIMENT NUMBER: 3

AIM: Create different network topologies using any network simulator (e.g. Cisco Packet Tracer, GNS3). For each topology, connectivity tests such as ping, file sharing between devices.

OBJECTIVES: To understand the concepts and characteristics of different network topologies such as Star, Bus, Ring, Mesh, and Tree using a network simulator.

THEORY:

Network Topology

A **network topology** is the physical or logical arrangement of computers, networking devices, and communication links in a network. It defines how devices are connected and how data flows between them. The choice of topology affects the network's performance, scalability, reliability, and maintenance.

Cisco Packet Tracer is a network simulation tool used to create and test different network topologies without requiring physical networking equipment. After designing the network, connectivity between devices can be verified using the **PING** utility.

1. Star Topology

In a **Star Topology**, all computers are connected to a central networking device such as a **switch** or **hub**. Communication between any two devices passes through the central device. It is the most widely used topology in modern Local Area Networks (LANs).

Advantages:

- Easy to install and manage.
- Failure of one cable affects only one device.
- Easy to add or remove devices.
- High network performance.

Limitations:

- Failure of the central switch disconnects the entire network.
- Requires more cabling than Bus topology.

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2. Bus Topology

In a **Bus Topology**, all computers are connected to a single communication cable called the **backbone cable**. Data travels along the backbone and is received by the intended device.

Advantages:

- Simple and inexpensive to install.
- Requires less cable.
- Suitable for small networks.

Limitations:

- Failure of the backbone cable stops the entire network.
- Difficult to troubleshoot.
- Performance decreases as more devices are added.

3. Ring Topology

In a **Ring Topology**, each device is connected to exactly two neighboring devices, forming a circular path. Data travels around the ring until it reaches the destination.

Advantages:

- Equal access for all devices.
- Better performance under heavy traffic.
- No data collisions.

Limitations:

- Failure of one device or cable can affect the whole network.
- Difficult to add or remove devices.

4. Mesh Topology

In a **Mesh Topology**, every device is directly connected to every other device. This provides multiple communication paths between devices.

Advantages:

- Very reliable and fault tolerant.
- Failure of one link does not interrupt communication.
- High security and data availability.

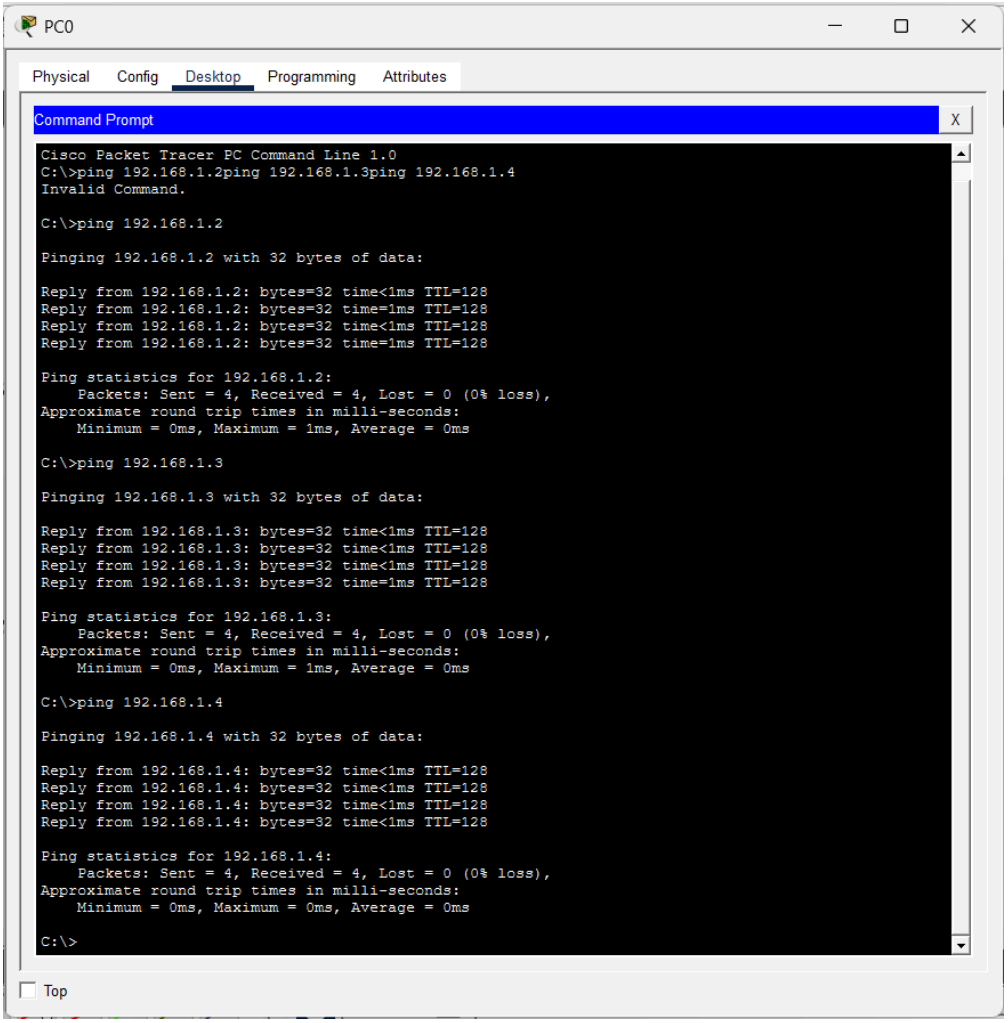
Limitations:

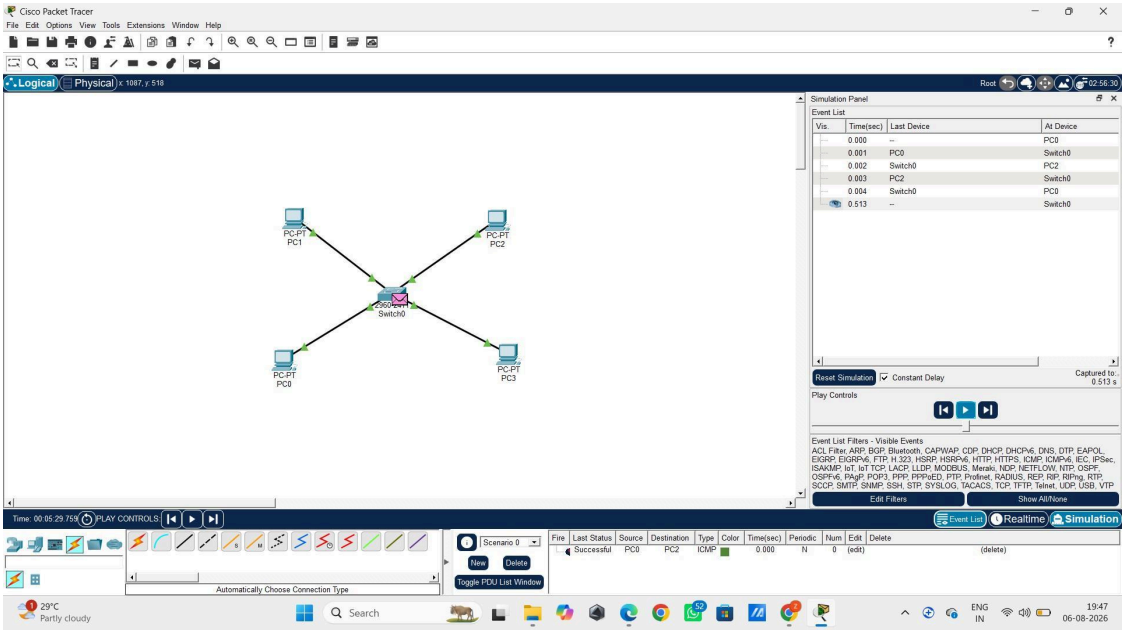
- Requires a large number of cables.
- Expensive to install and maintain.
- Complex network configuration.

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DIAGRAM:

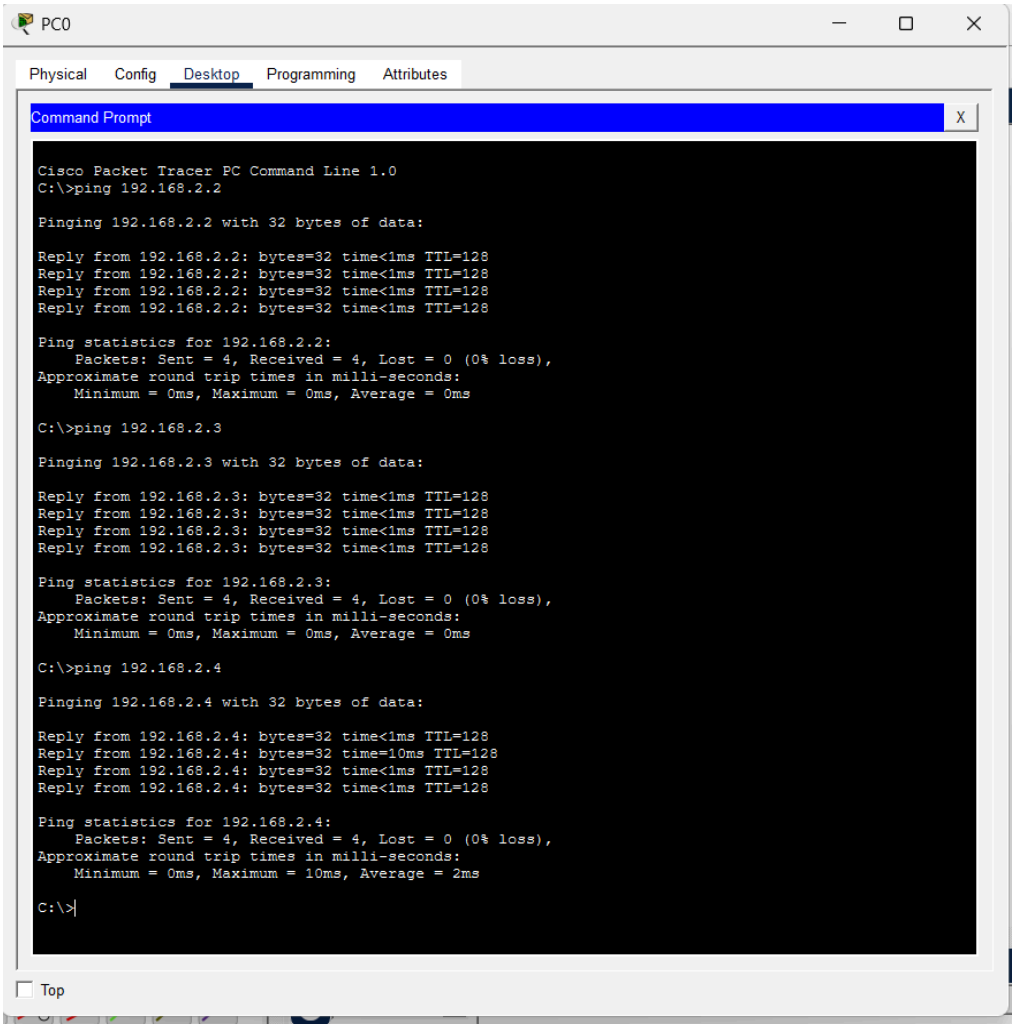
1. **Star topology**

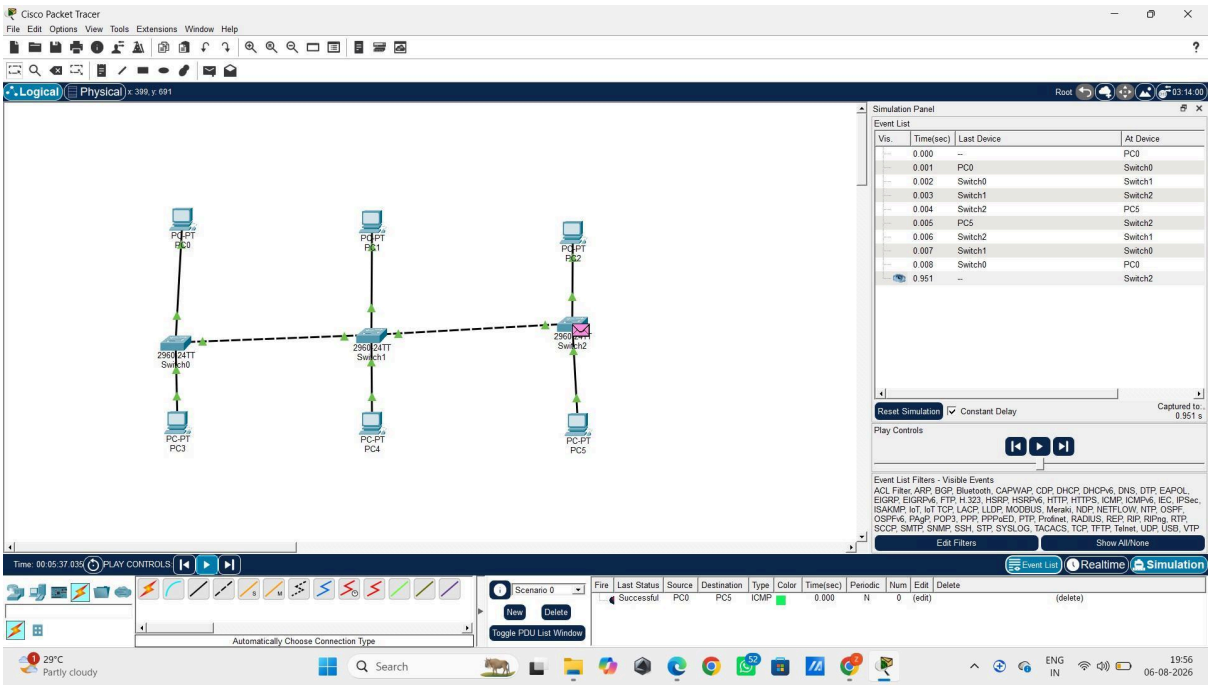




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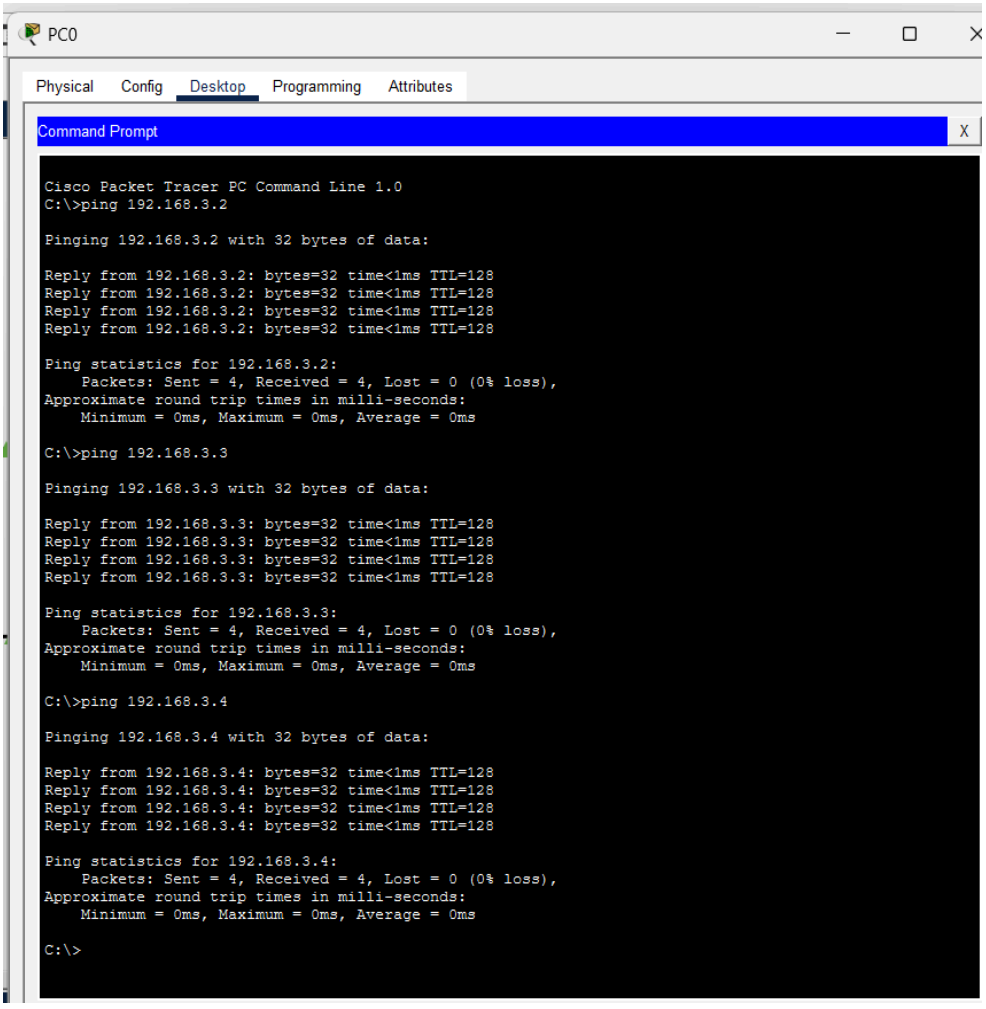
2. Bus topology

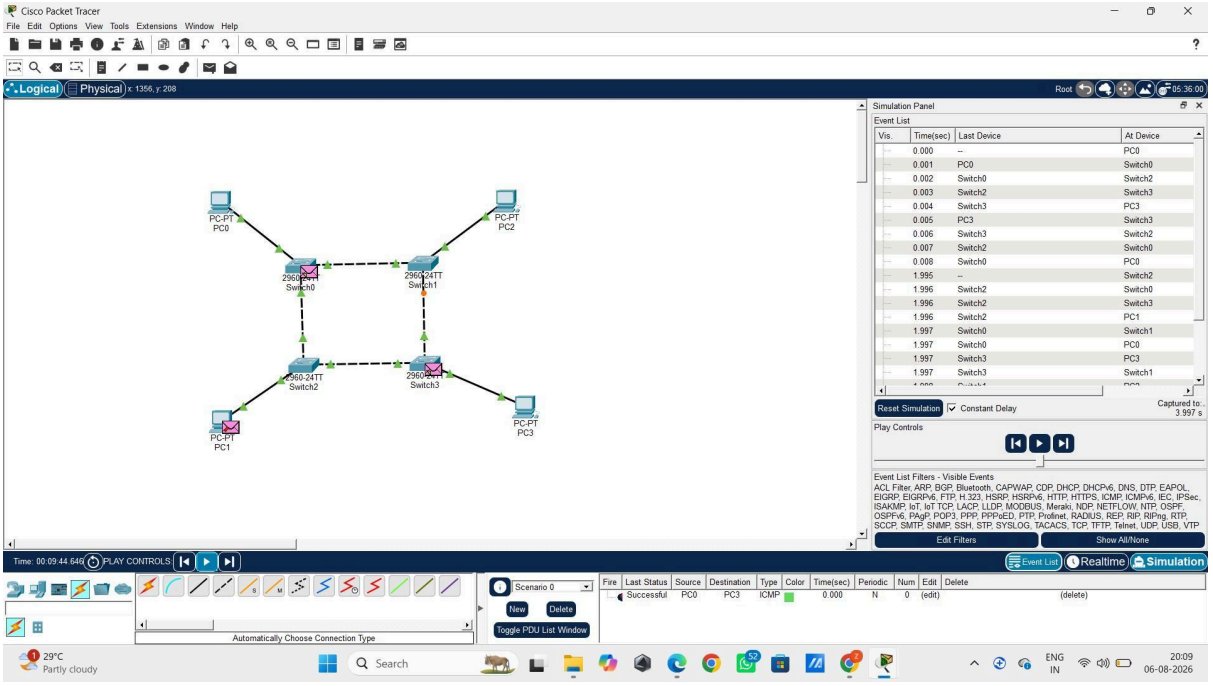




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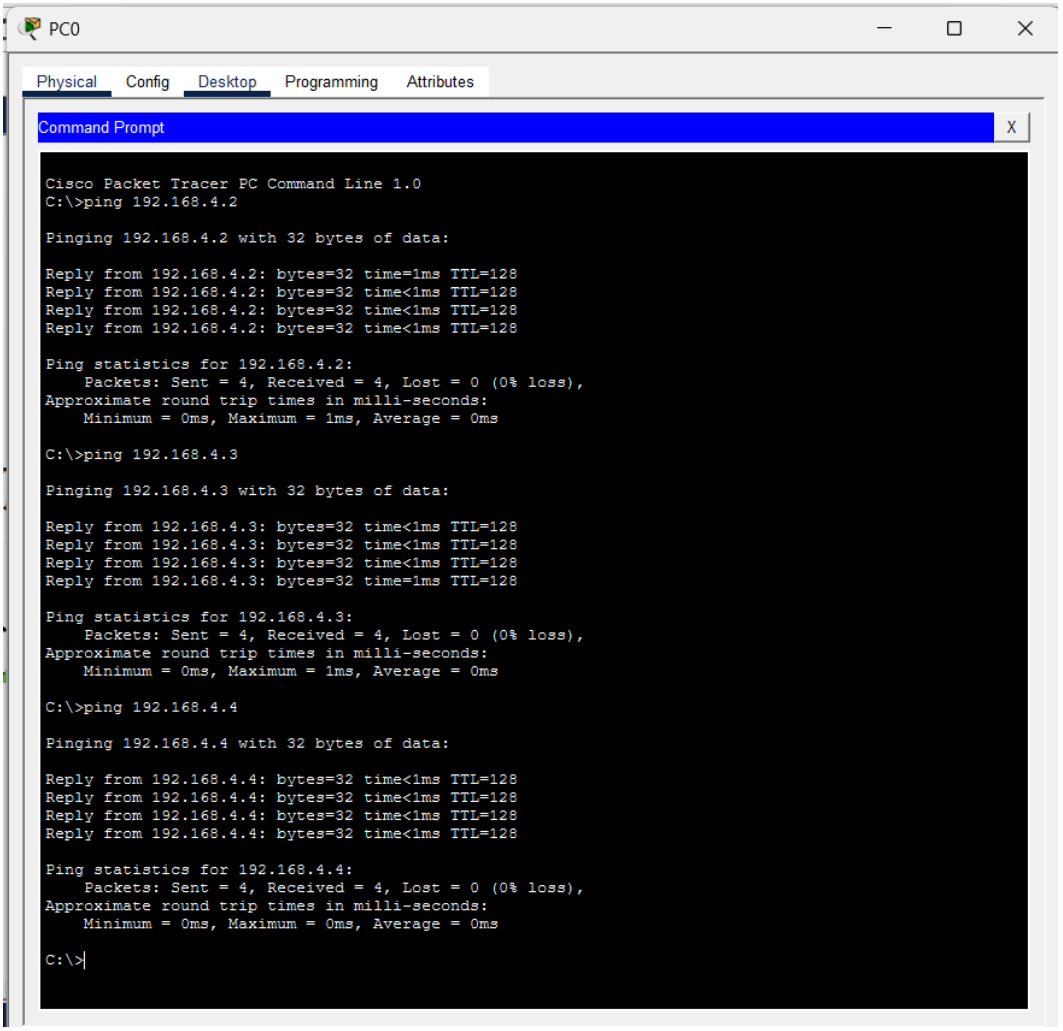
3. Ring topology

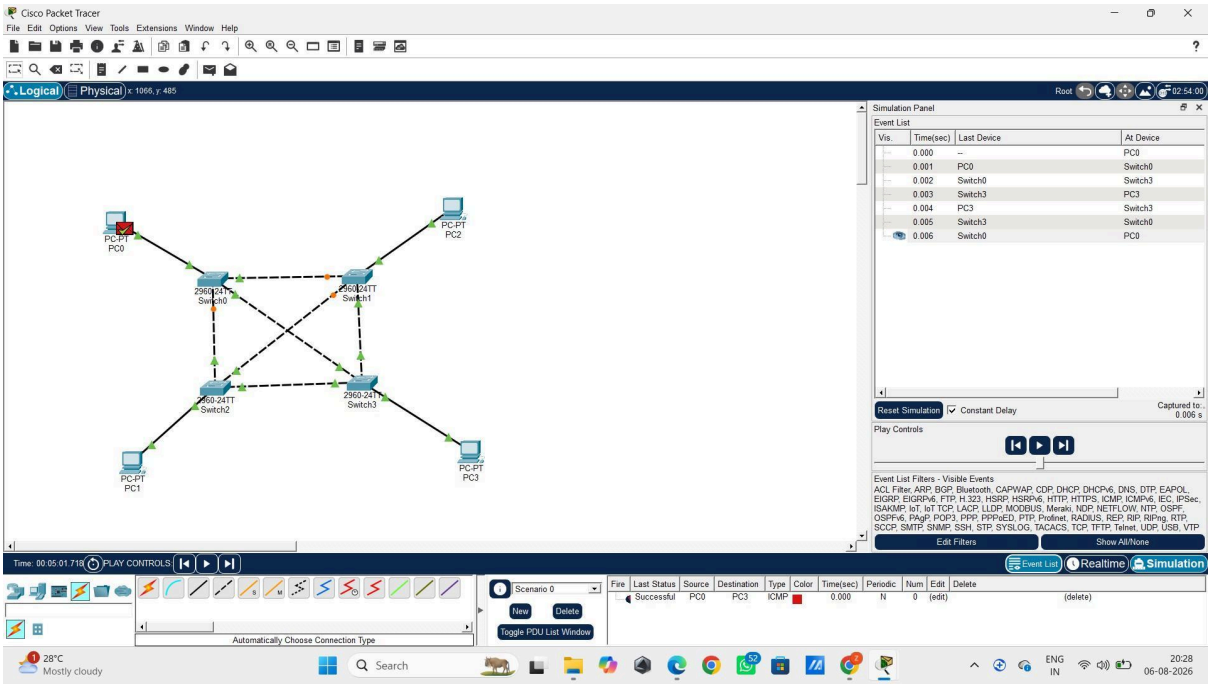




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4. Mesh topology





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RESOURCES REQUIRED:

- Cisco Packet Tracer Software
- Computer/Laptop
- Cisco 2960 Switch
- Hub
- Personal Computers (PCs)
- Copper Straight-Through Cables
- Ethernet Interfaces

PROCEDURES:

1. Open Cisco Packet Tracer on the computer.
2. Create a new project.
3. Add the required networking devices such as PCs, switches, routers, and hubs from the device list.
4. Design the required network topologies (Star, Bus, Ring and Mesh) by connecting the devices with appropriate cables.
5. Configure IPv4 addresses and subnet masks for all end devices.
6. Save each topology with a suitable name.
7. Use Simulation Mode to observe packet transmission between devices.
8. Verify network connectivity by performing the PING test between connected devices.
9. Test file sharing (where applicable) to ensure successful communication among devices.
10. Record the observations for each topology, including connectivity status, advantages, limitations, and any troubleshooting performed.

RESULT:

Different network topologies, namely **Star, Bus, Ring, and Mesh**, were successfully created using Cisco Packet Tracer. IPv4 addresses were configured correctly, and network connectivity was verified using the **PING** utility. All topologies successfully transmitted and received data packets, demonstrating proper communication between the connected devices.

CONCULSION:

The experiment was successfully performed by designing different network topologies in Cisco Packet Tracer. The characteristics, advantages, and limitations of **Star, Bus, Ring, and Mesh** topologies were studied. Network connectivity was verified using the **PING** utility, confirming successful communication between devices. This experiment provided practical knowledge of network design, topology implementation, IP configuration, and connectivity testing.